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Studierichting: Informatica
Oefeningenreeks 4A nr. 27.2

$$\int \frac{5xdx}{(x^2-3)^2}$$

$$u = x^2 - 3$$

$$du = 2xdx$$

$$xdx = \frac{du}{2}$$

$$\Rightarrow \int \frac{5}{(u)^2}\frac{du}{2}$$

$$= \frac{5}{2}\int \frac{1}{(u)^2}du$$

$$= \frac{5}{2}\int (u)^{-2}du$$

$$= \frac{5}{2} \cdot \frac{u^{-1}}{-1} + c$$

$$= \frac{5}{2} \cdot \frac{1}{-u} + c$$

$$= \frac{5}{-2u} + c$$

$$= \frac{5}{-2(x^2-3)} + c$$

References